DataNova: Automated EDA Report

Format: Presentation | Domain Detected: Healthcare

Executive / Dataset Overview - Healthcare-focused patient record set: 309 observations, 16 variables. - Target variable: LUNG_CANCER (binary label as string). - Demographic core: GENDER (categorical), AGE (numeric). - Lifestyle & symptom indicators: SMOKING, YELLOW_FINGERS, ANXIETY, PEER_PRESSURE, CHRONIC DISEASE, FATIGUE, ALLERGY, WHEEZING, ALCOHOL CONSUMING, COUGHING, SHORTNESS OF BREATH, SWALLOWING DIFFICULTY, CHEST PAIN. - All columns are fully populated; no missing entries detected. - Age distribution spans 21-88 years, with a median around the mid-50s (typical for lung-cancer screening cohorts).

Data Quality & Profiling Insights - Schema is clean: numeric fields are int64, categorical fields are strings. - No missing values or dtype mismatches - ready for modeling. - Outlier scan flags two low-age records (21 y, 38 y) below the lower bound of 39 y; these may represent early-onset cases or data entry errors. - Variable naming inconsistencies (trailing spaces in "FATIGUE " and "ALLERGY ") could cause column-reference bugs; recommend trimming whitespace. - Binary symptom fields are encoded as integers (0/1), suitable for logistic-type models. - Gender is the only non-numeric categorical field; consider one-hot encoding for algorithms that require numeric input.

Strategic / Operational Takeaways - The dataset is compact but well-structured; rapid prototyping of classification models (logistic regression, tree-based ensembles) is feasible. - Early-onset age outliers merit a focused clinical review - they could uncover high-risk sub-populations. - Clean, complete data eliminates the need for imputation, accelerating time-to-insight. - Standardizing column names (remove trailing spaces) will prevent downstream pipeline failures. - Because all symptom variables are binary, feature importance analysis will be straightforward, enabling clear communication of risk factors to clinicians. - With gender as the sole categorical predictor, stratified performance metrics (by sex) should be reported to ensure equitable model behavior. - The modest sample size (n = 309) suggests using cross-validation and possibly augmenting with external data for robust validation before deployment.

Visual Analysis